



## After a Decade of Genomic Testing, Are Milk Yields Being Impacted?

Jared Hutchins and Victor Funes-Leal

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Over the past 40 years, milk yield per cow has tripled in the United States. According to animal scientists at the Council of Dairy Cattle Breeding, at least half of this increase can be attributed to improvements in dairy cow genetics by selective breeding, with the other half explained by management and environmental factors. These breeding efforts in the United States impact dairy farmers in the United States and worldwide. In 2022, the United States exported 47% of the total value of global bull semen exports, making it the world's number one exporter of bull semen.

The market for cattle genetics has undergone significant changes. Around 2010, the first “genomically tested” bulls entered the dairy bull genetics market. Genomic testing is a tool for selective breeding that allows dairy breeders to estimate an animal's potential traits nearly at birth through a DNA sample. Before genomic testing, dairy bull breeders could only understand a bull's traits by waiting until it had offspring with data on milk yield and health outcomes. Cattle breeders can now speed up this process by estimating a bull's ability with a DNA test, allowing them to select bulls to sell more precisely and market them much more quickly.

Data on dairy bulls sold in the United States from 2000 to 2020 show several significant structural changes in the dairy genetics industry. Since genomic-proven bulls hit the market in 2010, dairy farmers have three times more bulls to choose from, and a bull's semen is now sold a little before they are 2 years old instead of at 5 years old. Genomic testing also coincides with a substantial increase in bull productive ability: Since 2010, the rate of growth in the predicted milk yield ability of dairy bulls has doubled. The percent of bulls on the market that are genomically tested has also risen to over 90%. Yet a puzzle remains: While genetic improvement has taken off, the rate of increase in the national average milk yield has scarcely changed since 2010 and has even decreased in recent years (USDA-NASS, 2024). Only states with smaller dairy herds (an average

herd size of fewer than 500 cows) have seen marginal increases in the rate of milk yield improvement, opening up the possibility that the new revolution in genetic improvement is not increasing on-farm milk yields across the board.

### The Supply of Dairy Bull Genetics

US dairy farmers predominantly use artificial insemination to breed their cows and raise future replacement cows. Instead of using a live bull, a farmer will buy semen from a bull sold by a genetics company or breeding cooperative. Farmers decide which bulls to breed with their cows by examining each bull's predicted genetic contribution to production and health outcomes and the physical characteristics of their offspring. These genetic traits include milk yield, pregnancy rate, milk cleanliness, and other health and production metrics. Three times a year, the organization Council on Dairy Cattle Breeding (CDCB) produces estimates of the contribution of each bull to its offspring's performance using data from dairy farms collected by Dairy Herd Improvement Associations (DHIAs) (Hutchins and Hueth, 2023). The CDCB publishes these estimates to help farmers understand how much, on average, the use of each bull will influence the characteristics of their future replacement.

Dairy breeders respond to farmers' demand for genetics by crossing female and male cows to produce bulls with desirable traits for the market. Once a bull is born, it takes at least 4 years to understand its traits with “daughter-proving,” meaning estimating a bull's traits using data on its daughters (see Box 1). A dairy bull becomes sexually mature at around 1 year of age. Once used on a dairy farm, its offspring will take at least 3 years to be born and reach producing age (10 months of gestation and 2 years from birth until a cow produces milk). Due to the time required to collect data, a bull is usually around 5 years old before it is advertised for sale. The age difference between a bull and its parents

## Box 1: How Is Genetic Potential Milk Yield Calculated for Dairy Bulls?

When choosing which bull to breed with, dairy farmers need to understand how each bull might impact the herd's milk production. They might also want to know how the bull's offspring will fare in terms of health and fertility. Using cow-level data from dairy herd improvement associations, the Council on Dairy Cattle Breeding produces bull genetic evaluations for traits like milk yield, which the industry can use freely. Each bull's genetic trait is calculated using a mixed linear model called the "animal model," which estimates the contribution of a dairy bull's genetics to its daughter's performance on multiple outcomes. For example, the milk yield genetic trait is calculated by measuring cow milk yield and determining how much variation in their milk yield can be explained by their father's identity. If all the daughters of one bull have high milk yields, even independent of other confounding factors, that bull will have a high "predicted transmitting ability" (PTA) for milk yield on the market. Each trait the CDCB calculates is a measure of a specific characteristic of the animal, such as productivity (e.g., milk, fat, protein yield), health (e.g., fertility, calving ability, predisposition to sickness), and physical characteristics (e.g., size of udders, foot health, body size).

is called the "generation interval" and directly influences how quickly dairy breeders can improve traits. The longer it takes to prove a genetic trait, the longer it will take for breeders to select and improve on that trait in the next generation.

In contrast to the daughter-proving method, genomic testing identifies genetic markers in each animal's DNA that correlate to desirable traits. Breeders can now take DNA samples from new animals and look for markers correlated with desirable traits. Using known correlations between genetic markers and production and health outcomes, a genomic test can estimate a dairy bull's genetic traits before it reaches sexual maturity.

Genomic testing drastically reduces the generation interval because bulls can be used to breed the next generation of cattle from an even earlier age. The accuracy of a genetic trait prediction is referred to as the "reliability" and is reported as a percentage, where 100% indicates certainty that the trait will be transmitted to its offspring. Before genomic testing, a bull with no offspring would have an average reliability rating of 28%–38% using only information on its parents. In contrast, a bull with a genomic test has an average reliability rating of 65%–75%, even with no information on its offspring. With the inclusion of data on the production of its offspring, reliability usually gets closer to 90%–95%. The accuracy of the genomic test for a bull without offspring is expected to continue to increase as more animals are genotyped and more genetic markers are found (Wiggans et al., 2011).

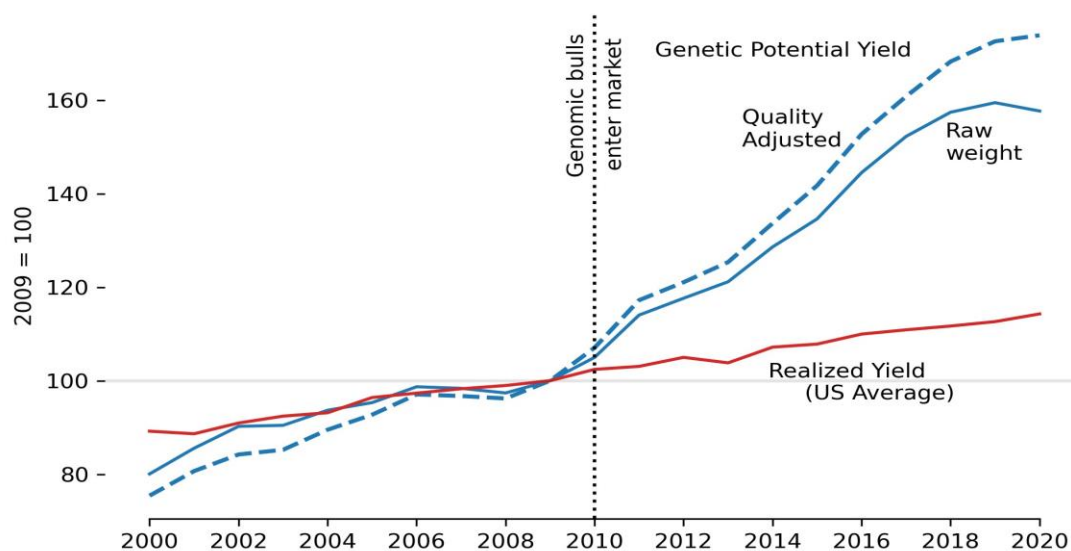
Genomic testing has had at least three significant impacts on the structure of the dairy bull genetics market. First, the age at which a bull is first marketed has dropped considerably, according to the National Association of Animal Breeders (NAAB), a trade organization that publishes a list of bulls available for purchase in North America three times a year. Before genomically tested bulls arrived, the average age was over 5 years. After the introduction of genomic testing, the average age has dropped to less than 2 years, a little

after a bull reaches sexual maturity. An accurate estimate of their traits without collecting data from offspring has allowed dairy breeders to market bulls nearly as soon as they can be marketed.

Second, according to the NAAB lists, the number of bulls available on the market has more than tripled. Starting in 2010, the number steadily increased from a little less than 500 before 2009 to more than 1,500 in 2020. This growth is entirely explained by genomic-proven bulls, who now account for the vast majority (over 90%) of the bulls on the market. Today, practically all bulls are genomically tested when they are born.

Third, the rate of change in genetic potential milk production, that is, how much more dairy bulls are predicted to increase milk yield, drastically increased after the introduction of genomic testing. This milk yield measure differs from the milk yield that dairy farmers see on their farms, as it is the milk yield change that the CDCB scientists predict will occur on average if a farm used that bull. For dairy bulls, the milk yield trait reported represents the genetic potential milk yield independent of other factors which can be more or less than the "realized" milk yield we would see on the farm or in aggregate data. In Figure 1, we can see the average genetic potential milk yield of dairy bulls for sale each year according to the NAAB, which measures the expected increase in pounds of milk yield from using each bull relative to bulls in the year 2000. The genetic potential yield for both raw weight and milk adjusted for the amount of protein and fat (standardized to 3.5% butterfat and 3.2% protein) increased over this period. Understanding raw weight and milk adjusted for components is especially important since most dairy farmers are paid based on components (butterfat, protein, and other milk solids) instead of raw weight. While the milk production ability of dairy bulls between 2000 and 2009 grew about 20%, it grew close to 60% from 2010 to 2020. Genetic potential for butterfat and protein also increased on the bull side. Genetic potential milk yield adjusted to have a uniform amount of fat and protein grew even more, almost 70% relative to 2009.

**Figure 1. US Average Milk Yield versus Genetic Milk Yield**



Genomic testing plays at least two significant roles in increasing genetic potential milk yield. The pace of change due to genetics is affected by both how selective breeders are about which bulls to use, referred to as “selection intensity,” and the time it takes to produce offspring from a match, which is the generation interval. Genomic testing allows dairy breeders to identify undesirable genetic lines earlier and remove them from the genetic pool without having to wait and measure the production of a bull’s offspring. The ability to be more selective at this stage amounts to an increase in selection intensity, which will cause the average genetic potential milk yield to increase more in each generation than before.

The average increase in genetic milk yield year-to-year also depends on the age of the bull when it produces offspring. The earlier a bull can be identified as productive, the younger it can be used to breed offspring. Thus, genomic testing decreases the generation interval, which causes genetic milk yield to increase more rapidly year after year. Combined, these two factors can help explain why the introduction of genomic testing caused an increase in the rate of change in traits dairy breeders find important.

While the adoption of genomically tested bulls at the farm level is not known, the percentage of genomically tested bulls on the market can be calculated from the NAAB bull lists. From these lists, we see that genomically-tested bulls quickly became the majority on the market. In 2010, the first year in which genomically tested bulls appear in the NAAB dairy bull lists, about 60% of the listed bulls were genomically tested. By 2015, 80% of the bulls on the list were genomically tested; by 2020, less than 5% of bulls were tested only using the daughter-proving method.

## Genetic Potential Milk Yield and Realized Milk Yield

The rate of change in milk yield predicted by genetics has more than doubled, but how did this impact the realized milk yields reflected in aggregate data? While dairy breeders focus on understanding the impact of genetics, milk yields are also impacted by management factors such as feeding practices and new technology and environmental factors such as changes in temperature and humidity. The CDCB estimates that about 50% of the growth in milk yields can be attributed just to genetics. Part of the remaining 50% is environmental factors such as high temperature and humidity, which can depress milk yields. On the management side, improvements in feed quality and milking technology are also important components of milk yield growth.

Figure 1 plots the average annual production of a dairy cow as measured by the milk production reports published by the USDA National Agricultural Statistics Service (NASS). In contrast to the CDCB’s measurement of genetic potential yield, this is the realized milk yield trend in the United States that reflects all factors, whether genetic or not. On the same graph, we plot genetic potential milk yield on the bull side and index both trends to 2009, 1 year before genomic bulls entered the market. Since our data only show the bull side, the improvements on the male side should represent no more than 25%–50% of the trend (bulls represent only half of the genetic trend, while female cows determine the other side). As the rate of genetic potential gain in milk yield sharply increased after 2009, we would expect a similarly noticeable increase in the US average milk yield trend.

| Table 1. Average Annual Percentage Growth |           |           |           |
|-------------------------------------------|-----------|-----------|-----------|
| Data Source                               | 2000–2020 | 2000–2009 | 2010–2020 |
| NASS                                      |           |           |           |
| Milk yield                                | 1.35      | 1.37      | 1.33      |
| Fat yield                                 | 1.71      | 1.34      | 2.00      |
| DHIA records                              |           |           |           |
| Milk yield                                | 0.71      | 0.48      | 0.89      |
| Fat yield                                 | 1.06      | 0.42      | 1.59      |
| Bull genetic estimates                    |           |           |           |
| Milk yield                                | 3.48      | 2.53      | 4.26      |
| Fat yield                                 | 5.04      | 4.08      | 5.83      |

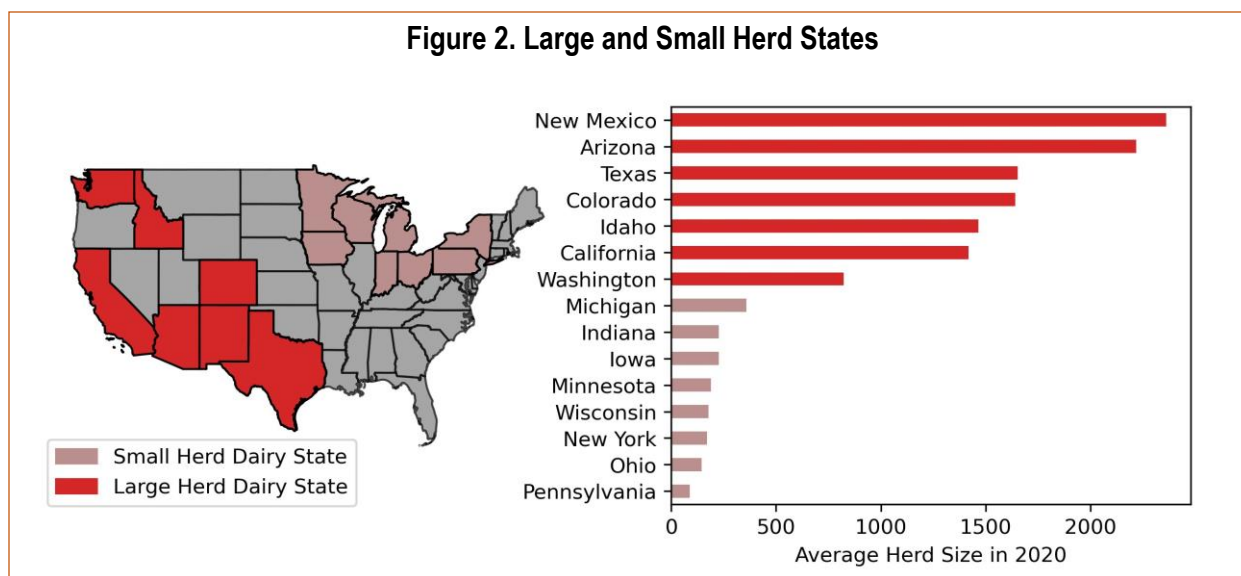
Instead, we see no noticeable change in the national average trend after 2009. From 2002 to 2009, the US milk yield growth rate closely followed the genetic milk yield growth rate. Both increased about 10% in those 7 years. After the introduction of genomic-proven bulls in 2010, genetic potential milk yield grew 60% in 10 years while the US milk yield trend grew at approximately the same rate. In the last 2 years, we even saw this rate slow: Milk annual production per cow increased only 0.1% between 2022 and 2023 (USDA, 2024).

Table 1 breaks down the average annual percentage growth in milk yield and butterfat yield across three data sources: NASS, DHIA, and bull genetic estimates. While no trend break is visible in the US average, bull genetic estimates are estimated using herds that are members of dairy herd improvement associations (DHIAs). If herds participating in DHIAs are more productive than the US average, bull genetic estimates may reflect genetic change on only DHIA farms and not farms on average.

Before 2010, the average US milk yield increased by an average 1.3% annually, while genetic milk yield increased on average 2.53% annually. After genomics, the average annual growth rate of US milk yield has

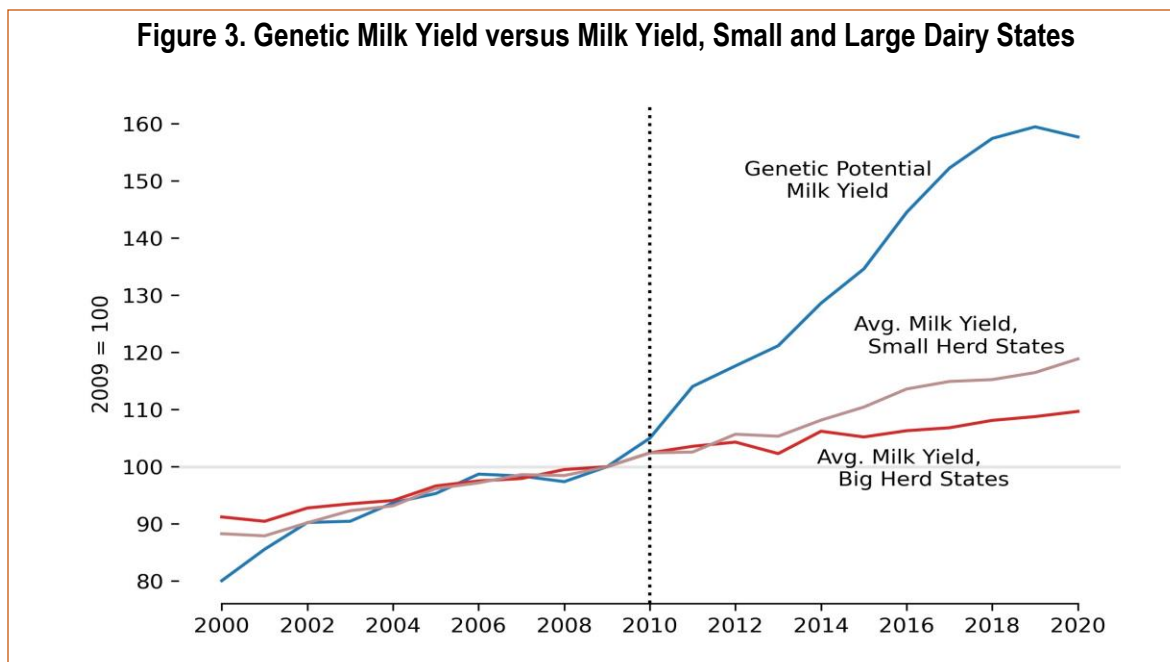
scarcely changed at 1.3% per year, while genetic milk yield has increased to 4.26%. While genomic testing appears to have caused a structural shift in genetic milk yield, US milk yield shows no such structural break. As measured on DHIA farms, milk yield appears to have increased rapidly over time but at a much lower rate (less than 1% growth a year).

To understand whether the trends in milk yields are similar across production systems, we break down milk yields in the 15 states with the most dairy cows with state-level data from NASS. We divide these states into “small herd states,” where the average herd size is less than 500, and “large herd states,” where the average herd size is more than 500. Figure 2 shows the small and large states and their average herd size in 2020. We use the 500-cow cutoff to divide states since it conveniently groups states geographically and is a good proxy for the different production scales in these two regions. Large dairy states are primarily in the Western United States and had an average herd size of more than 1,000 cows in 2020 (with the exception of Washington). Small dairy states are in the Midwest and Northeast and have more small herds, making the average herd size usually fewer than 300 cows.





**Figure 3. Genetic Milk Yield versus Milk Yield, Small and Large Dairy States**



In Figure 3, we chart average milk yields in these two categories of states. As in the aggregate, milk yield trends in both states follow the genetic milk yield trend quite closely before 2009. However, starting in 2013, the rate of milk yield growth in small states began to outpace large dairy states. Since new genetic improvements will take at least 3 years to affect on-farm averages, this is consistent with dairy farms in small-herd states benefiting more from genetic improvements in the genomics era than those in large-herd dairy states.

## Discussion

Why would the US milk yield average grow at the same rate while dairy bull genetics are predicted to deliver more milk yield than before? There are a number of possibilities. First, the adoption of higher-yielding genetics may not lead to higher milk yields on every farm. Some new bulls may produce more productive offspring in certain management environments or climates, a phenomenon quantitative geneticists refer to as “genotype-by-environment interactions.” These kinds of effects would lead to the efforts of breeders having different impacts in different circumstances, which may either enhance or depress the impact of genetic change on US milk yields. For example, farms that use sexed semen, which is almost guaranteed to produce female offspring, might choose dairy bulls that have higher productive ability since there is a higher chance what they chose will produce a replacement and not a male that will be sold for beef. One study on the adoption of dairy bulls in Wisconsin finds that a large portion of productivity can be explained by the type of herd in which the genetics are used (Hutchins et al., 2021). This might lead us to believe that smaller states have production systems where these genetics are the most productive, given differences in average farm size (Figure 3). Njuki (2022) also finds that medium-sized farms have productivity growth over the same period in

USDA data that is consistent with the use of higher-yielding genetics.

Nevertheless, there is little reliable data on the cost of dairy bull genetics, and this data gap makes it difficult to assess whether the genomics bulls available on the market are worth the investment for many farms. However, we also know that above 90% of dairy bulls in the market currently are genomically tested from birth, making it relatively unlikely that dairy farmers are failing to adopt any of these high-yielding bulls. Understanding how farms see the costs and benefits of these new bulls requires more data on the cost of these bulls and a more precise understanding of which farms adopt them.

Another possibility is that genetic estimates derived from DHIA farm data do not reflect the US average. In Table 1, we see that milk and fat yield change their growth rates in DHIA data, which is the data from which genetic estimates are derived. However, the growth rate is so small that it raises questions of whether the changes in trends match the changes in the genetic trend. In 2022, about 44% of US dairy cows had records in the DHIA system. However, a previous study found that farms on DHIA tend to be more productive, which may mean they are more likely to have management styles that help dairy bulls have higher yield impacts (Khanal et al., 2010).

A third possibility is that another factor has depressed yields even despite higher rates of genetic change. While dairy cattle can be protected from heat more easily than crops, high humidity, and high temperature combinations can harm dairy cow milk yields (Key et al., 2014; Hutchins et al., 2025 ). However, for this to explain the lack of change in milk yield from genetics, it would need to be the case that heat stress conditions dramatically worsened in 2010 to coincide with genomic testing just enough to counterbalance the increase in

genetic improvement. From what we know from studies of heat stress adaptation, dairy farms are becoming more resilient to heat stress, not less (Gisbert-Queral et al., 2021), making it unlikely that dairy farms became less resilient after 2010 to counter genetic change. Since average temperatures have risen gradually over time, we should have also seen the US average milk yield lag genetic change before 2010 instead of following it closely. While we do see the Midwestern states with lower summer temperatures have a higher rate of milk yield increase than the Western states, studies on heat stress losses on dairy farms find that the majority of losses are in Southern states (Key et al., 2014; Gisbert-Queral et al., 2021).

While we understand that genetic change has an

essential role in productivity growth, some particulars remain hazy. Future research in economics may be able to explain these trends by exploring how farmers choose different bulls and what sorts of farmers are likely to choose high-yielding bulls (Hutchins et al., 2021). In particular, there is little research on how much these new bulls cost and whether the cost is low enough to justify adoption by farmers. More work is also needed to understand how the increased frequency of heat waves will impact the adoption of certain dairy bulls and which production systems will see the largest returns to adopting high-yield bulls in a changing climate. Understanding exactly how dairy breeders' efforts contribute to increased milk yield in these production systems is crucial for the future of the US dairy sector and an important topic for research and discussion.

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## For More Information

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**About the Authors:** Jared Hutchins, Ph.D. ([jhtchns2@illinois.edu](mailto:jhtchns2@illinois.edu)) is an Assistant Professor with the Department of Agricultural and Consumer Economics at the University of Illinois Urbana-Champaign. Victor Funes Leal ([victorf2@illinois.edu](mailto:victorf2@illinois.edu)) is a Post-Doctoral Scholar with the Department of Agricultural Economics and Agribusiness at the University of Arkansas.